AI And Machine Learning

The Psychological Costs of Adopting AI

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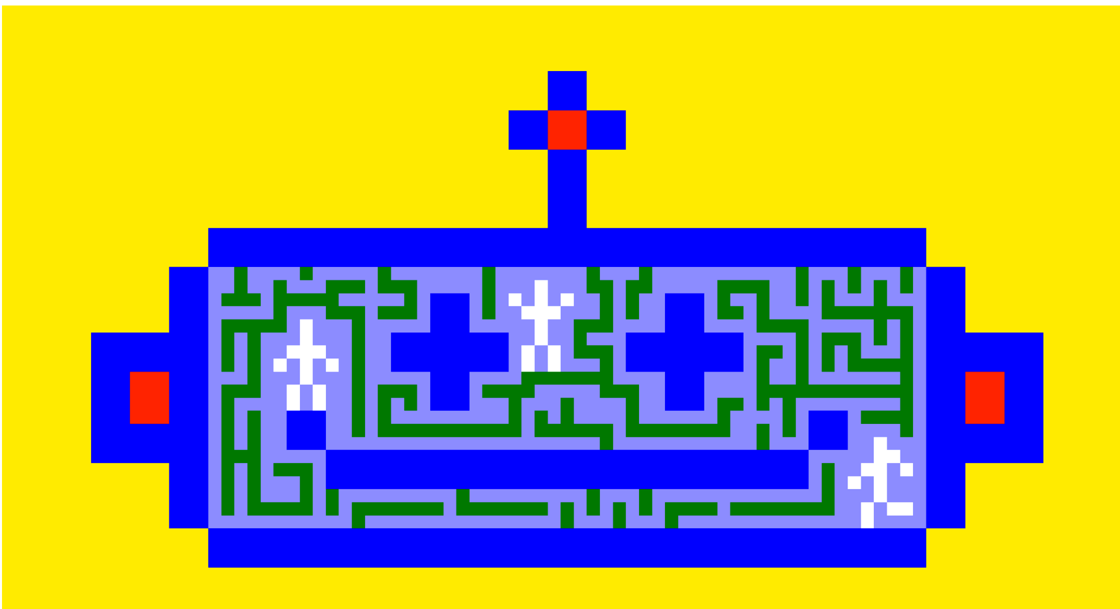

Illustration by Erik Carter

Summary. AI adoption strategies are overwhelmingly framed around productivity and efficiency. But that lens misses a critical constraint: the psychological cost of working with AI. New research shows that "psychological... **more**

As companies adopt AI, many are focused on improving efficiency and productivity—benefits advertised by the AI companies themselves. That focus might be a mistake. There is growing evidence that AI can have negative psychological consequences for employee motivation, which may even cancel out the benefits of AI, putting already embattled ROI models under even more scrutiny.

Leaders need to understand how AI use can degrade employee motivation, corrupt collaboration and innovation, and create higher levels of stress and burnout. All of these can raise serious barriers to successful AI adoption in organizations. The negative effects can, however, be significantly mitigated if companies take the psychological burden AI imposes into account in their adoption strategies. A number of leading-edge AI adopters have succeeded in doing that.

To explore how psychological debt may affect AI adoption, I surveyed more than 1,200 full-time employees in the U.S. and UK. Here's what leaders need to know.

Psychological Debt and Its Drivers

There are a number of potential negative psychological effects from the unstructured use of AI at work, all of which can impact motivation or behavior that's valuable for organizations. These have been identified either in recent AI adoption academic research or predate AI at work and instead focus on the drivers of motivation in the workplace. There are six forms that seem especially salient, collectively creating what is sometimes called psychological debt:

- **Cognitive debt.** Perhaps the most talked-about negative psychological consequence of unstructured AI use is the potential loss of cognitive processing and decision-making skills. For example, research suggests that the growing tendency to offload difficult tasks to AI (known as "cognitive offloading") undermines people's understanding of the problem and reduces the sense of ownership of the solution. Over time, this loss adds up, and cognitive debt grows.

- **Autonomy debt.** This describes the sense that AI is removing our ability to control how we work. Autonomy is recognized as a key source of motivation. Yet most conversations about AI optimization focus on workflow redesign, with little consideration of what that means for employees. Recent research has shown that even though business leaders are enthusiastically encouraging AI use among teams, when AI leadership focuses on productivity and technical expertise, employees relate AI to autonomy loss and replacement, leading to 'quiet quitting' and emotional tiredness.

- **Competency debt**. Competency describes how we are drawn towards tasks that present opportunities either for us to demonstrate or increase our expertise. In the context of AI, competency debt is the sense that the more we use AI for work, the less competent we become. Complex tasks that would ordinarily take us hours or even days are now completed in seconds and confer a sense of clarity and confidence that we often don't have in our own outputs. The less competent and confident we feel in our own abilities, the more likely we may become to rely on AI, exacerbating the challenge.

- **Relatedness debt**. Relatedness refers to our social instinct to be an integral part of social groups. Emerging research indicates, however, that increased AI usage decreases social interaction; AI provides clear and supportive responses, never argues, never tires, and has seemingly infinite patience. The leadership at one major UK university told me that this has led to students relying more and more on AI for study and learning —and made them less willing or able to engage in peer-led constructive argument formation and discussion. To mitigate that effect, I was told, the university is investing over £200 million to stimulate teamwork, discussion, and collaboration outside of the classroom.

- **Credibility debt.** A growing volume of research has documented how use of AI creates a sense that employees lose credibility in front of coworkers, even when those co-workers are also using AI in their roles. This would suggest that people find justification for own use but are critical of others doing the same thing. With senior employees having spent potentially decades building credibility in their professional environments, this potential loss of credibility could be highly damaging.
- **Identity debt**. Social identity theory posits that we belong to a host of social groups and, at any given moment, we identify with the group that gives us the clearest direction of how to behave in that moment. Where specific uses of AI are expected or mandated within certain workflows, professional social identity debt could accrue when these uses are seen as direct violations of group membership that damage the group. For example, creatives working within film, television and marketing services pride themselves on their ability to not only generate compelling ideas but to be able to do so almost to order; this is a defining quality of what it means to be a member of a distinctive social group. If they are then told to use AI to generate ideas or creative solutions, this will almost certainly be seen as an identity-damaging behavior, eroding one of the key drivers of distinctiveness for the group and those within it.

All six forms of psychological debt have the potential to cause damage to organizations pursuing AI integration strategies in a variety of ways: degrading employee motivation; corrupting collaboration and innovation efforts; and creating higher levels of stress and burnout. In addition, as employees look to avoid incurring these forms of psychological debt, they may resort to various coping behaviors, further denting productivity and efficiency, and requiring increases in training budgets.

How does psychological debt affect AI adoption?

To understand how psychological debt may affect AI adoption, I surveyed over 1,200 full time employees in organizations from 10 different sectors across the U.S. and UK as part of a research report on AI adoption I put together for Meltwater Consulting (of which I am a director). First, we measured employee reported levels of three important AI-related behaviors: the frequency of their AI use within their roles; the complexity of the tasks for which they used AI (from simple drafting exercises to strategic use cases); and their level of AI avoidance behavior (when they choose not to use AI in a specific task, even though they recognize it would improve outcomes).

We then measured levels of psychological debt via a series of statements and levels of agreement, then converted these scores into a 0–100 number (the larger the score, the higher the level of reported psychological debt). This enabled us to look both at overall levels of psychological debt and the specific dimensions of psychological debt, how they appear to impact adoption behaviors, and how they vary according to sector, function and role seniority.

We found that high levels of psychological debt were strongly related to:

How People Use AI

Psychological debt was almost twice as high (60) among those who reported using AI rarely versus people who used it multiple times a day (36). Moreover, people who only used AI for simple tasks had a psychological debt score of 46, whereas people who used it for complex and strategic tasks in their roles had a score of 35. At a more specific level, we found that professional social identity debt was strongly associated with variance in the complexity of tasks that AI is used for (the smaller the identity debt, the more complex the tasks) and in AI avoidance (the greater the debt, the more people avoided AI). A similar pattern emerged with relatedness debt: variations in debt are associated with significant variations in all three target adoption behaviors: when the debt is higher, frequency of AI use is lower, task complexity is lower, and AI avoidance behavior is higher.

Where They Are in Their Career

People at the very beginning of their career (up to five years in full-time employment) reported significantly higher psychological debt score (54) than those with more than 20 years' experience (40) and employees 10–20 years into their careers (44). In particular, more senior employees reported lower cognitive debt and professional social identity debt. This finding reflects the fact that people at the start of their careers feel a greater need to demonstrate their technical knowledge (at threat from AI), whereas those in senior roles have already demonstrated leadership skills (beyond the scope of AI). Interestingly, relatedness debt—the risk of being marginalized or less valuable to peers and working groups as a result of using AI—didn't change much between junior and senior roles. This may reflect a concern among senior managers that AI may diminish their role in creating and leading effective teams.

We also measured employee perceptions of AI relevance for their work. Over two thirds of employees found that AI was relevant to their work (70%) and nearly as many reported low levels of psychological debt. But only 41% of respondents reported both that AI was relevant and that they had low psychological debt. This suggests that more than half of those surveyed might benefit from training programs that combine building stronger business cases for the use of AI alongside systematically reducing or removing the risk of psychological debt.

So how might companies do that?

How to Shrink the Psychological Cost of Adopting AI

Reducing psychological debt will require that companies design human-AI relationships that recognize the sources of, and barriers to, human motivation and behavior. Fortunately, there is a wealth of research into human behavior that points to useful approaches that can be adapted to mitigate the negative effects documented:

Cognitive Debt

Cognitive debt can be avoided or paid down by ensuring that employees do not immediately turn to AI as a reflex action (cognitive surrender). This can be a significant temptation for employees as it provides what seems to be a fantastic (and fantastically easy) shortcut to a robust outcome in a fraction of the time. One way to preserve critical thinking involves building-in cognitive friction to the AI process, making its use less easy. For example, insisting that employees develop hypotheses or generate arguments before turning to AI ensures those higher-order cognitive functions stay active. For example, global financial services firm J.P. Morgan positions AI as an insights provider than a decision-maker. These insights and suggestions are useless unless they can be seen to fit into and support wider employee-generated arguments or hypotheses.

Autonomy Debt

Rather than spelling out the benefits of AI as a fait accompli, one recommendation is to work with employees to give them the information necessary to figure out for themselves when and how AI can help them. As part of their AI Principles in Practice program, the Netherlands-based banking group ING ensures that product teams must document how human-judgement is preserved before any model can be deployed, through discussion and collaboration with those who will be using the models. ING also works hard to ensure "explainability" for employees, describing models in plain language, explaining how models will work under specific conditions, and providing "nutrition labels" to show where data comes from, the limitations of the model, and known risks. This ensures employees can build their own arguments and viewpoints around the value of AI in their work.

Competency Debt

One way to remove or reduce the sense of diminishing competency in the face of AI is to ensure employees see AI as a form of support for their role rather than as a test of ability to continue in their role. For this to happen, a one-size-fits-all approach won't work, and AI training instead needs to recognize the specific contexts in which employees can and will turn to AI for support. Microsoft positioned itself as the first large enterprise to adopt Microsoft 365 Copilot and, to protect against the risk of competency debt, created their grassroots and peer-to-peer Copilot Champs Community initiative. This enabled employees to explore and understand the role relevance of AI in their work, in a trusted and dynamic environment, and in doing so helped everyone keep sight of their own skills and competencies in their roles.

Relatedness Debt

Protecting human-to-human collaboration in AI-enabled workflows is a key aspect of the human infrastructure needed in AI integration. This could involve designing processes where teams interpret AI outputs together, with cross-functional discussions, reviews and decision forums. These practices improve group dynamics and increase levels of trust among group members as they enable humans to display their human ability and expertise. For example, research has shown that when consumer packaged goods P&G coordinates cross-functional teams to collectively review innovation outputs from AI chatbots, not only does performance increase, but normally siloed roles become more collaborative in the process.

Credibility Debt

Shadow AI use can be seen as an attempt to avoid accruing credibility debt—if no one knows you are using AI, there cannot be any risk to your credibility. One solution is to bring AI use out of the shadows, making its use visible, shared and institutional. This has the potential to re-set a corporate culture around AI-supportive behaviors. For example, the fintech firm Klarna worked hard to make its Kiki AI assistant not just a tool but a cultural device within the business. Introduced in June 2023, the business announced 90% of employees were using Kiki within a year, and Kiki had answered over 250,000 employee questions.

Communicating the scale and frequency of use can reduce credibility debt as it represents a social norm within the culture of the business. Interestingly, Klarna CEO explained that experimentation is the norm within the company, saying “We push everyone to test, test, test, and explore.” Communicating this norm further strengthens the use case of Kiki (as the key means of testing and exploring), which in turn removes the risk of credibility debt for employees.

Professional Identity Debt

Debt of this kind can be reduced by establishing explicit identity-consistent norms for AI use – just moving when AI is used within a process can sometimes be enough. For example, it has been well-documented that medical clinicians have pushed back on the introduction of AI in their roles, specifically because of the potential loss of identity in terms of expertise, accuracy and patient care. To mitigate this problem, medical equipment manufacturer Philips has looked to position AI use as an identity-affirming behavior, drawing attention how AI use increases precision (strengthening consultant diagnoses and prognoses), removes logistical bottlenecks (and so freeing-up consultants to focus on exercising their expertise) and supports multidisciplinary collaboration in cancer care (where expertise becomes more visible and valuable). Philips has worked to identify specific actions or moments when AI can be valuable and in doing so further enhance the social identity of a clinician.

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It is impossible to predict at this point just how AI will transform the workplace. However, one thing is certain: understanding and building the right human infrastructure will be as important as picking the right AI tools. Former U.S. Surgeon General, C. Everett Koop famously pointed out that "drugs don't work in patients that don't take them." Both obvious and profound, the same can be said of gen AI; even the best tools won't work in organizations whose employees won't use them. The organizations that reap the greatest rewards from AI will be those that recognize that adoption is not just a technological challenge, but also a psychological one.

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